DO NOW — RECALL

Complexity and algorithm choice

Name: _____ Class: _____ Date: _____

1. Why is a queue suitable for a print server?

2. What condition must the array satisfy?

3. How many times does a for-loop from i=0 to i<6 execute?

4. A loop of length 4 inside another loop of length 4: how many inner operations?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It processes jobs in arrival order (FIFO).

2. Model answer

It must be sorted.

3. Model answer

Six times.

4. Model answer

16.

Vocabulary — know these terms

Big O — Asymptotic growth of algorithm work.

O(1) — Constant growth.

O(log n) — Logarithmic growth, e.g. halving search intervals.

O(n) — Linear growth.

O(n^2) — Quadratic growth, e.g. complete nested scans.

NEW LEARNING

Read, annotate and explain

Start with n

n is the number of items. More items usually mean more operations.

Trace loop nesting

One full scan performs n steps. A nested n-by-n scan performs n^2 steps.

Compare growth

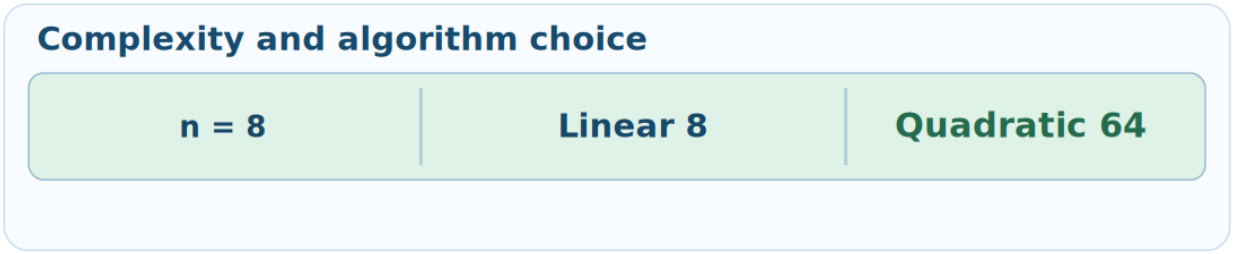
When n doubles, linear work doubles; quadratic work roughly quadruples. Big O describes growth, not seconds.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

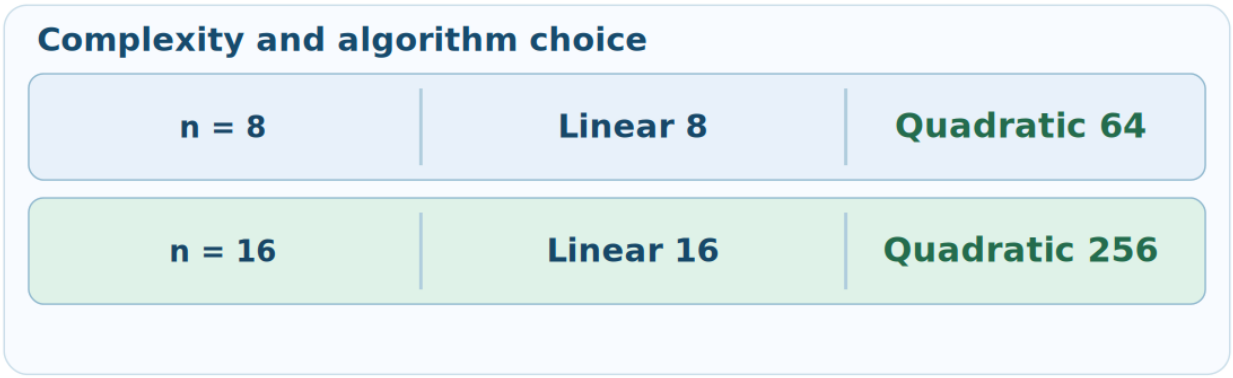
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Complexity and algorithm choice: n = 8, Linear 8, Quadratic 64

New Learning

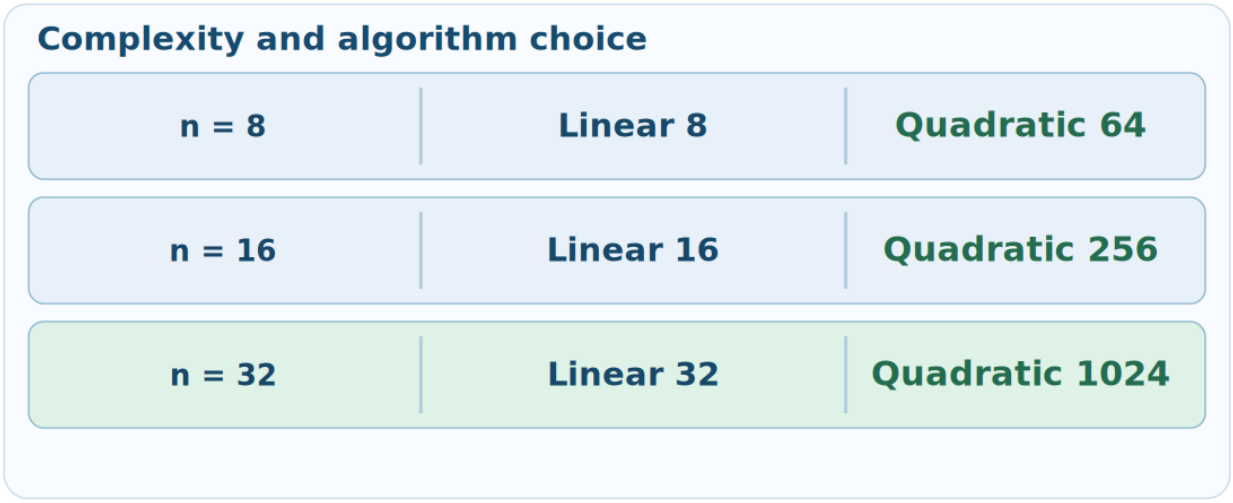


Complexity and algorithm choice: n = 8, Linear 8, Quadratic 64; n = 16, Linear 16, Quadratic 256

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Complexity and algorithm choice: n = 8, Linear 8, Quadratic 64; n = 16, Linear 16, Quadratic 256; n = 32, Linear 32, Quadratic 1024

I DO — WORKED EXAMPLE

Follow the worked method

Compare operation counts of one loop and two nested loops for n=5.

Predict

One loop visits five elements, so does five operations.

Explain

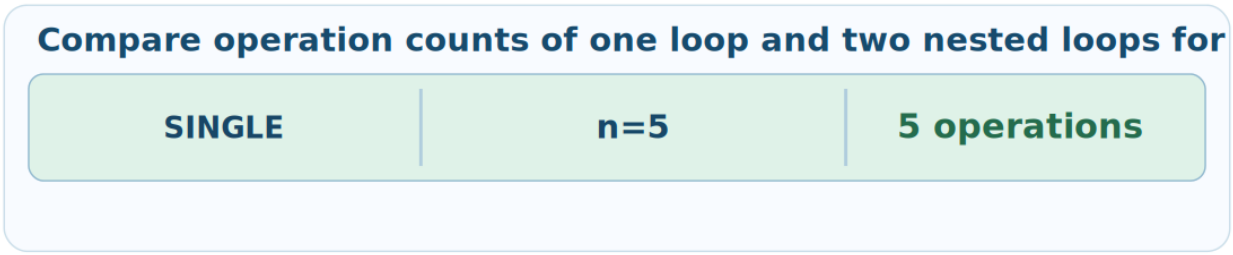
The nested loop repeats five operations five times, making 25.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Compare operation counts of one loop and two nested loops for n=5.

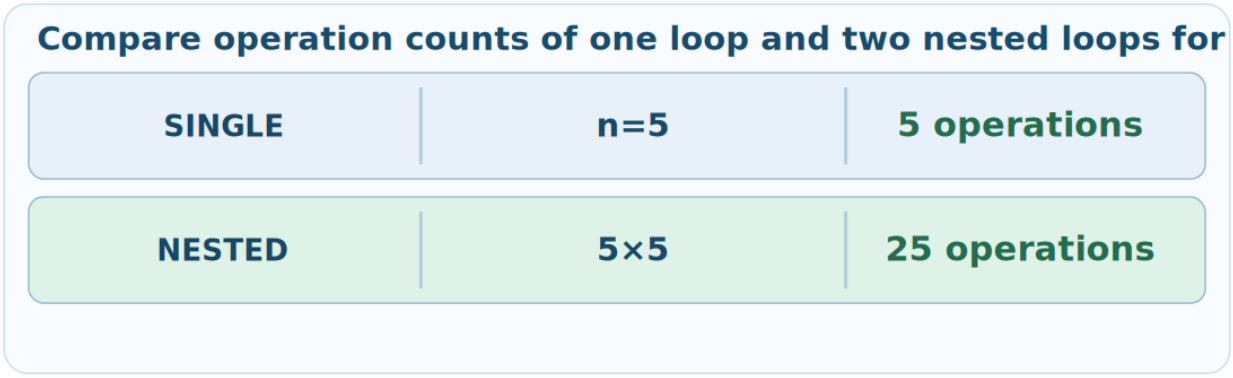
One loop visits five elements, so does five operations.



Compare operation counts of one loop and two nested loops for n=5.: SINGLE, n=5, 5 operations

Compare operation counts of one loop and two nested loops for n=5.

One loop visits five elements, so does five operations.



Compare operation counts of one loop and two nested loops for n=5.: SINGLE, n=5, 5 operations; NESTED, 5×5, 25 operations

WE DO — GUIDED PRACTICE

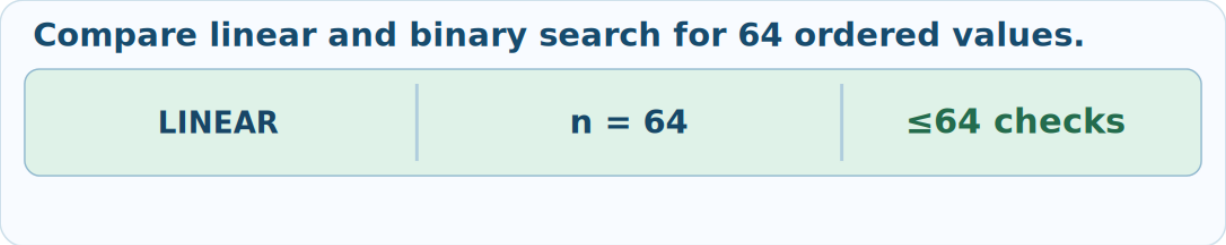
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Compare linear and binary search for 64 ordered values.

Linear search can check all 64 values.



Compare linear and binary search for 64 ordered values.: LINEAR, n = 64, ≤64 checks

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict Linear search can check all 64 values.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give the Big O of a single full scan of n elements.

Question 2

For n=100, give iterations of a linear n loop and nested n×n loop.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why may linear search be preferable for a single search of a short unsorted array?

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

$O(n^2)$ means an algorithm takes n^2 seconds.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

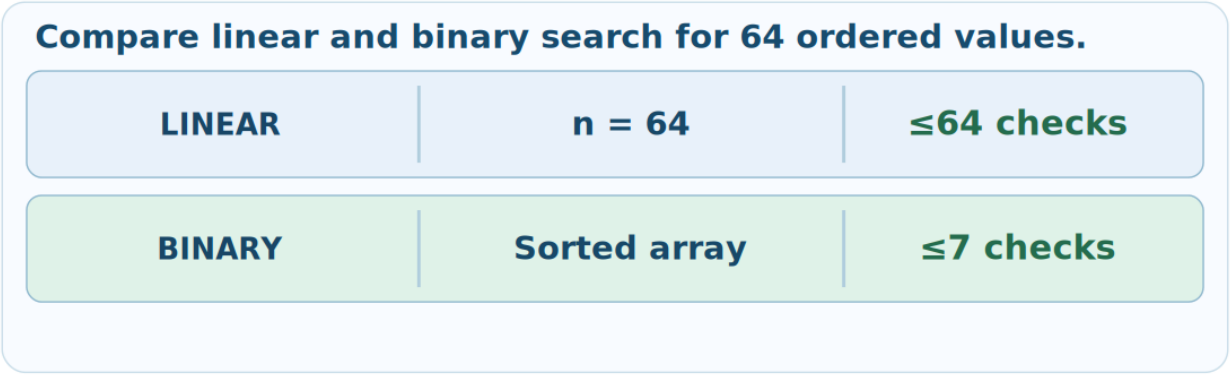
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Compare linear and binary search for 64 ordered values.

Linear search can check all 64 values.



Compare linear and binary search for 64 ordered values.: LINEAR, n = 64, ≤64 checks; BINARY, Sorted array, ≤7 checks

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain

A standard midpoint binary search needs at most seven comparisons on 64 sorted values.

IMPROVE: Explain one correction to your own reasoning.